

# Quicksort’s Lognormal Fit and Non-Normal Limit

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## Abstract

Best-case, worst-case, and average-case analyses compress an algorithm’s behavior to a few numbers. Yet different inputs produce a distribution of costs, and that distribution has a shape even when its mean is known. We examine the number of comparisons made by classical randomized-pivot Quicksort. Two seemingly conflicting facts hold. For every  $n \geq 4$ , the best ordinary lognormal fit has a strictly higher expected log-density than the best normal fit. Nevertheless, a fitted lognormal does not approach Quicksort’s standardized limiting distribution: the latter retains skewness 0.854881 . . . , while the lognormal becomes asymptotically normal after standardization. We give a moment-based, computer-assisted proof of the finite-size inequality and a short argument for the limiting separation. The main text develops the questions and their interpretation; proofs and exact certificate details follow in the appendix.

**Keywords:** Quicksort; comparison count; runtime distribution; lognormal fit; expected log score.

## 1 Beyond three numbers of complexity

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An algorithm does not have a single running cost at a fixed input size. It has a collection of possible costs, one for each input and, for randomized algorithms, each sequence of random choices. Best-case and worst-case bounds describe the extremes. Average-case analysis describes a center once a probability model has been chosen. None of these numbers, by itself, says how common the costs between the extremes are.

That missing shape can answer natural questions. Is a run moderately above average unusual? Is the long-cost side heavier than the short-cost side? If two algorithms have the same expected cost, does one vary more? Such questions call for the *distribution* of cost over the input and randomness space. The distribution contains the familiar summaries, but also records their arrangement: spread, asymmetry, and tails.

There are two ways to study that shape. At a given input size, one can ask which simple probability family describes the costs better. As the size grows, one can ask whether that family becomes the *true limiting shape* after removing the changing center and scale. A fit can win the first comparison without winning the second. Quicksort offers a clean example of this distinction.

## 2 A distribution generated by recursive splits

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We count comparisons, rather than wall-clock time, to isolate the algorithm from machine effects. Consider  $n$  distinct keys. At each recursive call, choose a pivot uniformly among the keys in that subproblem, compare it with the other keys, and recurse on the two sides. Let  $C_n$  be the total count. With  $C_0 = C_1 = 0$ , its distribution satisfies, for  $n \geq 2$ ,

$$C_n \stackrel{d}{=} n - 1 + C_{I_n} + C'_{n-1-I_n}, \quad I_n \sim \text{Unif}\{0, \dots, n-1\}, \quad (1)$$

Here  $\stackrel{d}{=}$  means equality in distribution,  $I_n$  is uniform on the displayed integers, and the prime denotes an independent recursive copy. The two recursive counts are independent conditional on  $I_n$ . Equation (1) describes an entire distribution, not just an expected runtime. Balanced splits tend to save comparisons; repeated uneven splits increase the count.

For a small illustration, the full distribution at  $n = 4$  is

| comparisons $c$            | 4   | 5   | 6   |
|----------------------------|-----|-----|-----|
| probability $\Pr(C_4 = c)$ | 1/2 | 1/6 | 1/3 |

Its best case is 4, worst case is 6, and mean is  $29/6$ . The table says more: the middle cost is the least common of the three. For large  $n$ , the support and probabilities are far richer, but the same recurrence determines them.

Write  $\mu_n = \mathbb{E}C_n$ ,  $v_n = \text{Var}(C_n)$ , and  $m_{r,n} = \mathbb{E}(C_n - \mu_n)^r$  for the mean, variance, and  $r$ th central moment;  $\mathbb{E}$  and  $\text{Var}$  denote expectation and variance. Every  $C_n$  with  $n \geq 2$  is strictly positive. Standard identities give [3, 2]

$$\mu_n = 2(n+1)H_n - 4n, \quad (2)$$

$$v_n = 7n^2 + 13n - 2(n+1)H_n - 4(n+1)^2 H_n^{(2)}, \quad (3)$$

where  $H_n^{(r)} = \sum_{j=1}^n j^{-r}$  and  $H_n = H_n^{(1)}$ . All logarithms below are natural unless a base is shown.

The mean grows like  $n \log n$ , while the standard deviation grows like  $n$ . Relative fluctuations shrink, even though the number of comparisons still varies substantially on the  $n$  scale. This observation will matter for both results.

### 3 Result 1: a lognormal wins the finite-size fit

A histogram of comparison counts can look somewhat bell-shaped, but with a longer right side. It is therefore natural to compare a normal density with an ordinary lognormal density; the latter is the distribution of  $e^G$  for a Gaussian variable  $G$ . We make the phrase *better fit* precise: for each family, choose the parameters maximizing the *expected log-density at the actual comparison count*. This is a population score under the exact distribution of  $C_n$ , not a judgment by eye or a fit to a simulated sample.

Let  $S_N(X)$  and  $S_L(X)$  denote the optimal expected log-densities for normal and ordinary lognormal families when  $X > 0$ . The normal optimum matches the mean and variance of  $X$ ; the lognormal optimum matches the mean and variance of  $\log X$  in log space. The exact score difference is

$$\Delta(X) = S_L(X) - S_N(X) = \frac{1}{2} \log \frac{\text{Var } X}{\text{Var}(\log X)} - \mathbb{E} \log X. \quad (4)$$

Positive  $\Delta$  means that the lognormal assigns a higher expected log-density to the observed counts. Counts are discrete and these models continuous; evaluating both densities at the same count is a comparative score, not a claim that either density is the exact probability mass function.

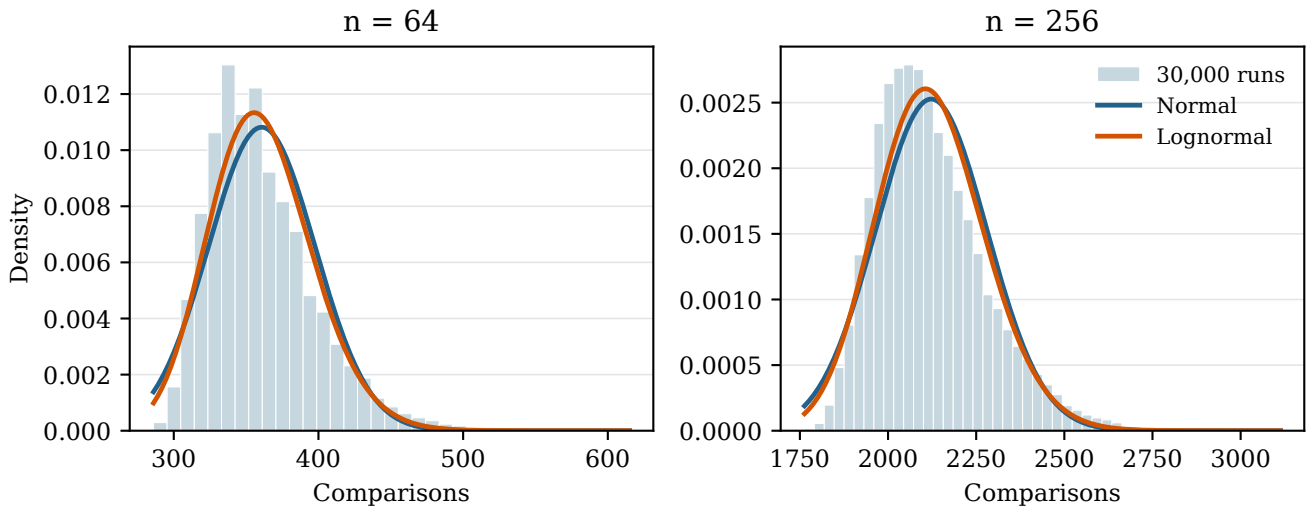


Figure 1: Comparison-count histograms from 30,000 independent randomized-pivot runs at each displayed size. The normal and ordinary lognormal curves maximize the log-density on each simulated sample. The sample mean log-density favors the lognormal by 0.0420 at  $n = 64$  and 0.0279 at  $n = 256$ ; the all-size claim below concerns the exact population score.

$$\Delta(C_n) > 0 \quad \text{for every integer } n \geq 4. \quad (5)$$

In plain terms, this comparison has no later crossover: under this particular scoring rule, the best lognormal beats the best normal at every size from four onward. The statement is stronger than a histogram comparison at a few sizes. It has a specific scope: ordinary lognormal and normal densities optimized by expected log-density. A different criterion or a shifted lognormal is a different question.

The proof turns the fit comparison into a condition on the third and fourth central moments. A useful general criterion is

$$\min X \geq \frac{2}{3}\mathbb{E}X \quad \text{and} \quad 3(\mathbb{E}X)\mathbb{E}(X - \mathbb{E}X)^3 > 4\mathbb{E}(X - \mathbb{E}X)^4 \implies \Delta(X) > 0. \quad (6)$$

Here  $\min X$  is the smallest value in the support of  $X$  (its essential infimum in general). The third-moment term captures rightward asymmetry; the fourth moment controls the error in approximating the logarithm. For Quicksort, exact rational interval checks establish both conditions for  $4 \leq n < 10,000$ . Uniform analytic bounds establish them for every  $n \geq 10,000$ . Appendix A provides the proof and certificate details. The calculation uses published Quicksort moment identities [3]; the all-size fit inequality is obtained by combining them with (6).

## 4 Result 2: the fitted shape does not become exact

Winning a finite-size score comparison does not mean that Quicksort is becoming lognormal. To ask about its eventual shape, first remove the moving mean and scale:

$$\hat{C}_n = \frac{C_n - \mu_n}{\sqrt{v_n}}. \quad (7)$$

The classical Quicksort limit theorem says that  $\hat{C}_n$  converges in distribution to a non-normal random variable  $Q$  [1, 2]. In particular, the known moment formulas give its skewness as

$$\text{skew}(Q) = \frac{16\zeta(3) - 19}{(7 - 2\pi^2/3)^{3/2}} = 0.854881867\dots \quad (8)$$

Here  $\text{skew}(W) = \mathbb{E}(W - \mathbb{E}W)^3 / (\text{Var } W)^{3/2}$  for a variable  $W$  with positive variance, and  $\zeta(r) = \sum_{j=1}^{\infty} j^{-r}$  for  $r > 1$ . The positive constant says that the long right side survives the passage to large inputs.

Now let  $L_n$  be an ordinary lognormal with the *same mean and variance* as  $C_n$ . Its relative spread shrinks:

$$\frac{\sqrt{v_n}}{\mu_n} = \Theta(1/\log n) \longrightarrow 0. \quad (9)$$

The  $\Theta$  notation means that the ratio of the two positive quantities stays between fixed positive constants for all sufficiently large  $n$ . When a lognormal becomes narrow relative to its mean, its centered and standardized shape tends to a normal. Its skewness therefore tends to zero, while Quicksort's tends to the nonzero constant above. More precisely, with  $\hat{L}_n = (L_n - \mu_n)/\sqrt{v_n}$ ,

$$\hat{C}_n \Rightarrow Q, \quad \hat{L}_n \Rightarrow \mathcal{N}(0, 1). \quad (10)$$

The arrow  $\Rightarrow$  denotes convergence in distribution, and  $\mathcal{N}(0, 1)$  is the normal law with mean zero and variance one. The largest gap between their cumulative distribution functions (CDFs) has a positive limit:

$$\lim_{n \rightarrow \infty} \sup_x |\Pr(C_n \leq x) - \Pr(L_n \leq x)| = d_K(\mathcal{L}(Q), \mathcal{N}(0, 1)) > 0. \quad (11)$$

Here  $\mathcal{L}(Q)$  is the law of  $Q$ , and  $d_K(A, B)$  is the supremum of the absolute difference between the CDFs of laws  $A$  and  $B$ ; it is unchanged by applying the same increasing affine transformation to both laws. This is a precise sense in which the lognormal approximation never converges to the true standardized Quicksort shape. The log-score-optimal ordinary lognormal from Result 1 also has an asymptotically normal standardized shape. Appendix B proves these statements.

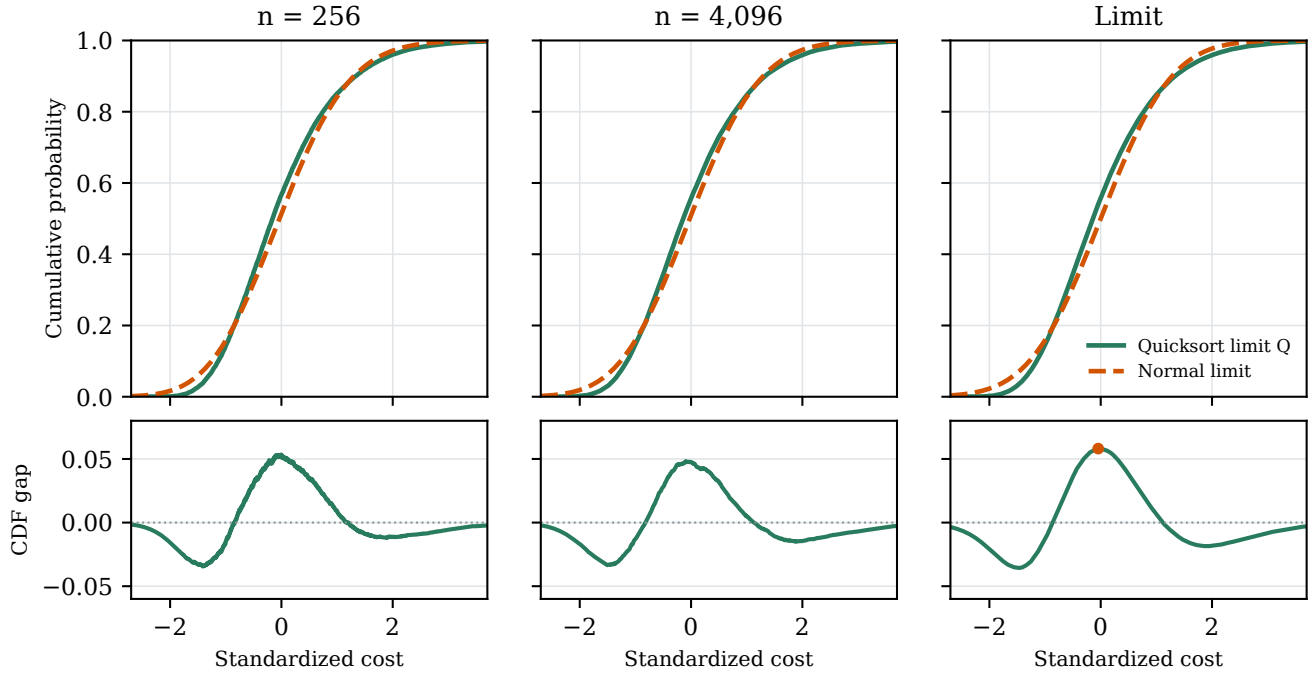


Figure 2: Standardized CDFs at two finite sizes and in the limit; the lower row shows Quicksort minus fitted-model CDF. Each finite Quicksort curve uses 30,000 independent runs, and each dashed curve is the ordinary lognormal with the exact Quicksort mean and variance. The limiting Quicksort curve approximates  $Q$  by Monte Carlo iteration of its recursive fixed-point equation; its dashed counterpart is the standard normal CDF. The estimated limiting maximum gap is 0.058, and Equation (11) proves it remains positive.

## 5 Putting the two statements together

The results describe two scales of one phenomenon. On the scale of the raw positive count, a mild right asymmetry helps a lognormal beat a symmetric normal in expected log-density. On the scale of fluctuations around the mean, random recursive splits keep an asymmetric limiting shape. A fitted lognormal becomes approximately normal on that standardized scale because its relative spread shrinks.

This distinction is useful beyond Quicksort. When an algorithm's cost is summarized by a fitted curve, ask what the fitting rule measures and whether that curve also describes the cost after centering and scaling. A good score at every finite size and an incorrect limiting law can coexist. These results apply to the uniform-pivot comparison-count model in (1); other implementations and costs require their own analysis.

## Appendix A Proof of the finite-size fit result

For a positive random variable  $X$ , let  $S_N(X)$  and  $S_L(X)$  be the suprema of  $\mathbb{E} \log f(X)$  over ordinary normal densities on  $\mathbb{R}$  and ordinary lognormal densities on  $(0, \infty)$ , respectively. The normal optimizers are  $\mathbb{E}X, \text{Var } X$ ; the lognormal's log-location and log-variance optimizers are  $\mathbb{E} \log X, \text{Var}(\log X)$ . Direct substitution gives the exact score gap

$$\Delta(X) := S_L(X) - S_N(X) = \frac{1}{2} \log \frac{\text{Var } X}{\text{Var}(\log X)} - \mathbb{E} \log X. \quad (12)$$

The quantity is dimensionless: changing the unit of  $X$  cancels between the two terms.

**Lemma 1** (Moment criterion). *Let  $X > 0$ ,  $\mu = \mathbb{E}X$ ,  $v = \text{Var } X$ ,  $m_r = \mathbb{E}(X - \mu)^r$ , and  $\min X \geq 2\mu/3$ . If  $3\mu m_3 > 4m_4$ , then  $\Delta(X) > 0$ .*

*Proof.* Put  $Z = (X - \mu)/\mu \geq -1/3$ . The elementary inequality

$$\log^2(1+z) \leq z^2 - z^3 + \frac{4}{3}z^4 \quad (z \geq -1/3) \quad (13)$$

implies  $\text{Var}(\log X) \leq \mathbb{E} \log^2(1+Z) < \mathbb{E} Z^2 = v/\mu^2$ . Also  $\mathbb{E} \log(1+Z) < \log(1+\mathbb{E} Z) = 0$  by strict Jensen. Replacing  $\text{Var}(\log X)$  in (12) by its strict upper bound shows  $\Delta(X) > 0$ .

For completeness, (13) follows by expanding  $\log^2(1+z) = \sum_{k \geq 2} (-1)^k (2H_{k-1}/k) z^k$  for  $|z| < 1$ . For  $0 \leq z \leq 1$  the alternating-series bound through degree four is  $z^2 - z^3 + (11/12)z^4$ . For  $z \geq 1$ , use  $\log^2(1+z) \leq z^2 \leq z^2 - z^3 + 4z^4/3$ . For  $z = -y$  with  $0 \leq y \leq 1/3$ , the coefficients from degree five onward are at most  $5/6$ , so their sum is bounded by  $(5/6)y^5/(1-y) \leq (5/12)y^4$ . Adding the degree-four coefficient  $11/12$  gives  $4/3$ .  $\square$

**Theorem 1** (Strict log-score advantage). *For the count in (1),  $S_L(C_n) > S_N(C_n)$  for every integer  $n \geq 4$ .*

*Proof.* We certify the two hypotheses of Lemma 1. A balanced recursion tree yields the exact minimum

$$b_n = (n+1)h - 2^{h+1} + 2, \quad h = \lfloor \log_2 n \rfloor. \quad (14)$$

The published third and fourth central-moment formulas [3, Thms. 2.3-2.4] determine  $m_{3,n}$  and  $m_{4,n}$  from  $H_n^{(1)}, \dots, H_n^{(4)}$ . Their third-moment expression, for example, is

$$\begin{aligned} m_{3,n} = & -n(19n^2 + 81n + 104) + 14(n+1)H_n \\ & + 12(n+1)^2 H_n^{(2)} + 16(n+1)^3 H_n^{(3)}. \end{aligned} \quad (15)$$

For the 9,996 sizes  $4 \leq n < 10,000$ , the accompanying exact-rational certificate evaluates these formulas by outward intervals and verifies

$$3b_n - 2\mu_n > 0, \quad 3\mu_n m_{3,n} - 4m_{4,n} > 0. \quad (16)$$

No simulations or floating-point sign tests enter the assertions. The computation independently checks the moment identities against the full recurrence (1) for  $n \leq 12$ . The certificate procedure is summarized below.

For  $n \geq N = 10,000$ , the same inequalities admit uniform analytic bounds. Set  $t = N^{-1}$  and  $u = 1+t$ . The harmonic tail estimates

$$\zeta(r) - \frac{1}{(r-1)n^{r-1}} < H_n^{(r)} < \zeta(r) \quad (r > 1), \quad (17)$$

and  $H_n \leq \log n + 1$  imply  $H_n/n < 11/N$ : the function  $(\log x + 1)/x$  decreases for  $x > 1$  and  $\log N < 10$ . From (15), dropping positive terms where helpful, one gets

$$\frac{m_{3,n}}{n^3} > 16\zeta(3) - 19 - 81t - 104t^2 - 8u^3t^2 > \frac{11}{50}. \quad (18)$$

Substitution into the fourth-moment formula and removal of its remaining negative terms gives

$$\begin{aligned} \frac{m_{4,n}}{n^4} < & \frac{2260}{9} + \frac{9658}{9}t + \frac{15497}{9}t^2 + \frac{11357}{9}t^3 - 168(\zeta(2) - t) \\ & + 12u^2(11t)^2 + 48u^3(11t)\zeta(2) + 48u^4\zeta(2)^2 \\ & - 96\left(\zeta(4) - \frac{t^3}{3}\right) < \frac{11}{10}. \end{aligned} \quad (19)$$

The last numerical inequalities use rational enclosing bounds  $1.6449 < \zeta(2) < 1.645$ ,  $\zeta(3) > 1.202$ , and  $\zeta(4) > 1.0823$ , each certified by integral remainders after 200 terms. Further,  $\mu_n > 14n$ , since  $H_n > \log n > 9$  for  $n \geq N$ ; hence  $3\mu_n m_{3,n} > 3(14n)(11n^3/50) > 4(11n^4/10) > 4m_{4,n}$ .

Finally, (14) gives  $b_n/n \geq \log_2 n - 2$ . Together with  $H_n \leq \log n + 1$  and  $n \geq N$ , this implies  $b_n \geq 2\mu_n/3$ : after rearrangement it suffices that

$$\left(\frac{1}{\log 2} - \frac{4}{3}\right) \log n \geq \frac{2}{3} + \frac{4(\log n + 1)}{3n}, \quad (20)$$

which holds at  $N$  and strengthens as  $n$  grows. Thus Lemma 1 applies on the entire tail as well.  $\square$

## Appendix B Proof of the limiting separation

The advantage above might suggest that Quicksort becomes lognormal at large  $n$ . Its standardized limit says otherwise. Let  $L_n$  be the ordinary lognormal with the same mean  $\mu_n$  and variance  $v_n$  as  $C_n$ , and put

$$\hat{C}_n = (C_n - \mu_n)/\sqrt{v_n}, \quad \hat{L}_n = (L_n - \mu_n)/\sqrt{v_n}.$$

**Theorem 2** (Asymptotic separation). *The variables  $\hat{C}_n$  converge to Quicksort's standardized limit  $Q$ , whereas  $\hat{L}_n \Rightarrow \mathcal{N}(0, 1)$ . In particular,*

$$\lim_{n \rightarrow \infty} \text{skew}(C_n) = \frac{16\zeta(3) - 19}{(7 - 2\pi^2/3)^{3/2}} = 0.854881867\dots, \quad (21)$$

while  $\text{skew}(L_n) \rightarrow 0$ . Moreover, for the CDFs,

$$\lim_{n \rightarrow \infty} \sup_x |\Pr(C_n \leq x) - \Pr(L_n \leq x)| = d_K(\mathcal{L}(Q), \mathcal{N}(0, 1)) > 0. \quad (22)$$

The same normal limiting shape holds for the ordinary lognormal optimized in Theorem 1, after standardization by its own mean and standard deviation.

*Proof.* The Quicksort limit theorem establishes  $(C_n - \mu_n)/n \Rightarrow Y$  with a continuous limiting distribution [1, 2];  $v_n/n^2 \rightarrow \sigma^2 = 7 - 2\pi^2/3 > 0$  from (3). Its limiting third central moment follows from (15):  $m_{3,n}/n^3 \rightarrow 16\zeta(3) - 19$ . Consequently  $Q = Y/\sigma$  has the nonzero skewness in (21); see also [3]. The notation  $a_n \sim b_n$  below means  $a_n/b_n \rightarrow 1$ , and  $a_n = O(b_n)$  means  $|a_n|/b_n$  stays bounded for positive  $b_n$ .

For the limiting curve in Figure 2, we approximate  $Y$  using its recursive characterization [1]:

$$Y \stackrel{d}{=} UY_1 + (1 - U)Y_2 + 1 + 2U \log U + 2(1 - U) \log(1 - U),$$

where  $U$  is uniform on  $(0, 1)$  and  $Y_1, Y_2$  are independent copies of  $Y$ , independent of  $U$ . The plotted variable is  $Q = Y/\sigma$ ; the accompanying `make_figures.py` uses 25 population iterations of 400,000 values.

For the moment-matched lognormal, its underlying Gaussian log-variance is  $s_n^2 = \log(1 + v_n/\mu_n^2) = O(1/\log^2 n)$ , since  $\mu_n \sim 2n \log n$  and  $v_n \sim \sigma^2 n^2$ . Hence  $s_n \rightarrow 0$  and

$$\hat{L}_n = \frac{\exp(s_n Z - s_n^2/2) - 1}{\sqrt{\exp(s_n^2) - 1}} \Rightarrow Z, \quad Z \sim \mathcal{N}(0, 1).$$

Its skewness  $(e^{s_n^2} + 2)\sqrt{e^{s_n^2} - 1}$  therefore vanishes. Nonzero skewness proves  $Q \neq \mathcal{N}(0, 1)$ . Continuity of both limit CDFs makes each Kolmogorov convergence uniform, proving (22) by the triangle inequality.

For the log-score-optimal lognormal, the minimum bound in the preceding proof and the fourth-moment bounds yield a Taylor expansion of  $\log C_n$  around  $\mu_n$ :  $\text{Var}(\log C_n) \sim v_n/\mu_n^2 \rightarrow 0$ . Its log-variance thus also tends to zero, giving the same standardized normal limit.  $\square$

## Appendix C Exact certificate specification

This appendix records enough detail to reproduce the finite interval computation without sampling. Define  $H_r = H_n^{(r)}$  and  $a = n + 1$ . The fourth central moment used in the test is the published formula [3]

$$\begin{aligned} m_{4,n} = & \frac{n(2260n^3 + 9658n^2 + 15497n + 11357)}{9} \\ & - 2a(42n^2 + 78n + 77)H_1 + 12a^2H_1^2 \\ & + [-4a^2(42n^2 + 78n + 31) + 48a^3H_1]H_2 \\ & - 96a^3H_3 + 48a^4H_2^2 - 96a^4H_4. \end{aligned}$$

For each  $n = 4, \dots, 9999$ , maintain four integer sums

$$S_r(n) = \sum_{j=1}^n \lfloor D/j^r \rfloor, \quad D = 10^{24}, \quad r = 1, 2, 3, 4.$$

Substitute  $[S_r(n)/D, (S_r(n) + n)/D]$  for  $H_r$  in (2), (15), and the expression above. Ordinary outward interval addition and multiplication produce intervals for  $\mu_n, m_{3,n}, m_{4,n}$ . Assert that the lower endpoints of  $3b_n - 2\mu_n$  and  $3\mu_n m_{3,n} - 4m_{4,n}$  are strictly positive. The integer floor bound gives an interval containing each harmonic number, so a passing lower endpoint proves the corresponding exact inequality. The certificate reports 9,996 successful sizes; direct probability mass functions generated from (1) independently cross-check the moment formulas for  $2 \leq n \leq 12$ .

For the tail constants, set  $M = 200$  and  $T_r = \sum_{j=1}^M j^{-r}$ . Integral comparison supplies rational enclosures

$$T_r + \frac{1}{(r-1)(M+1)^{r-1}} < \zeta(r) < T_r + \frac{1}{(r-1)M^{r-1}} \quad (r = 2, 3, 4).$$

They imply  $1.6449 < \zeta(2) < 1.645$ ,  $\zeta(3) > 1.202$ , and  $\zeta(4) > 1.0823$ . Substituting these rational values into the right sides of (18) and (19), with  $t = 10^{-4}$ , yields respectively  $> 0.2238$  and  $< 1.0194$ . These estimates are deliberately looser than necessary; the displayed proof only needs 0.22 and 1.1. In particular, the  $\log N < 10$  observation controls  $H_n/n$  through monotonicity of  $(\log n + 1)/n$ ; it is *not* applied as the false statement  $\log n < 10$  for all  $n \geq N$ .

The minimum comparison formula (14) follows from splitting as evenly as possible at every node. If  $n = 2^h q$  with  $1 \leq q < 2$ , then  $b_n/n \geq h - 2/q$ . Since  $\log_2 q + 2/q \leq 2$  on  $[1, 2]$ , this implies  $b_n/n \geq \log_2 n - 2$ . This final elementary inequality makes the analytic tail independent of the finite certificate.

## References

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- [1] U. Rösler. A limit theorem for “Quicksort.” *RAIRO Theoretical Informatics and Applications*, 25(1):85–100, 1991. [https://www.numdam.org/item/ITA\\_1991\\_\\_25\\_1\\_85\\_0/](https://www.numdam.org/item/ITA_1991__25_1_85_0/).
- [2] J. A. Fill and S. Janson. Quicksort asymptotics. *Journal of Algorithms*, 44(1):4–28, 2002. <https://arxiv.org/abs/math/0105248>.
- [3] Y. Yao. A detailed analysis of Quicksort algorithms with experimental mathematics. Manuscript, 2019. <https://sites.math.rutgers.edu/~yao/QuickSort.pdf>.